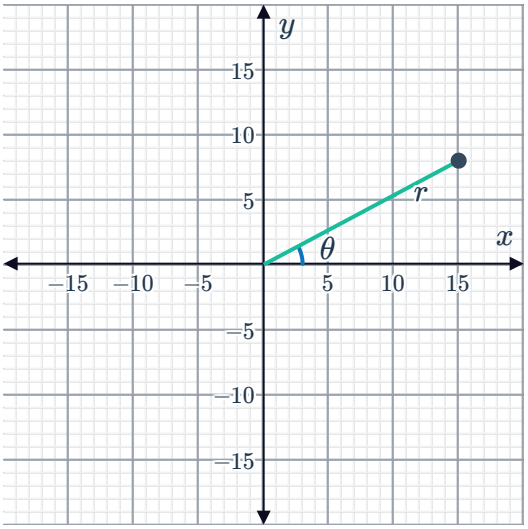
Trigonometric ratios from points

1 The point on the following graph has coordinates $(15, 8)$.



a Find r, the distance from the point to the origin.

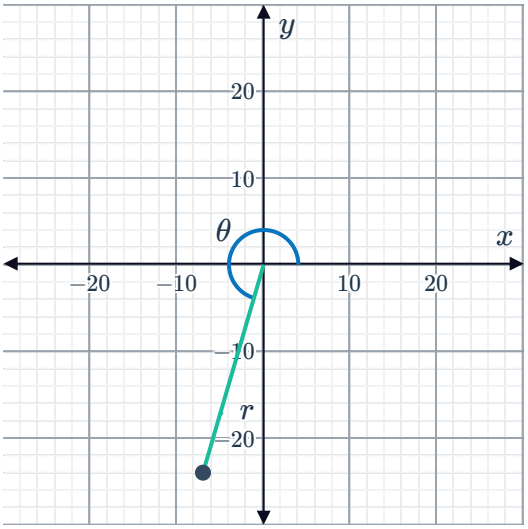
b Find the value of the following:

i $\sin\theta$ **ii** $\cos\theta$

iii $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

2 The point on the following graph has coordinates $(-7, -24)$.



a Find r, the distance from the point to the origin.

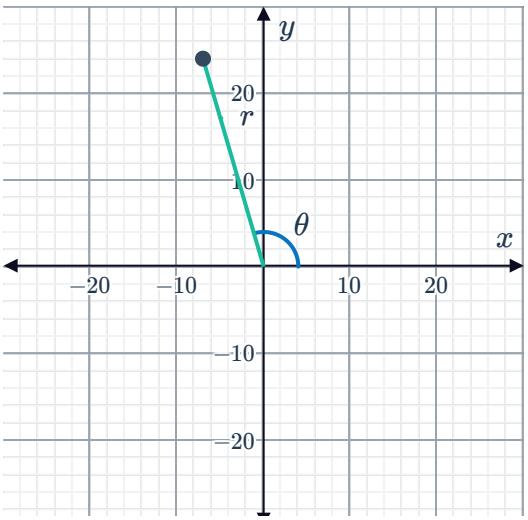
b Find the value of the following:

i $\sin\theta$ **ii** $\cos\theta$

iii $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

3 The point on the following graph has coordinates $(-7, 24)$.



a Find r, the distance from the point to the origin.

b Find the value of the following:

i $\sin\theta$ **ii** $\cos\theta$

iii $\tan\theta$ **iv** $\csc\theta$

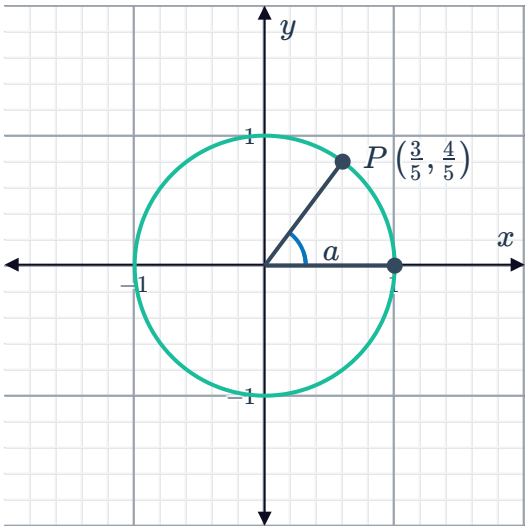
v $\sec\theta$ **vi** $\cot\theta$

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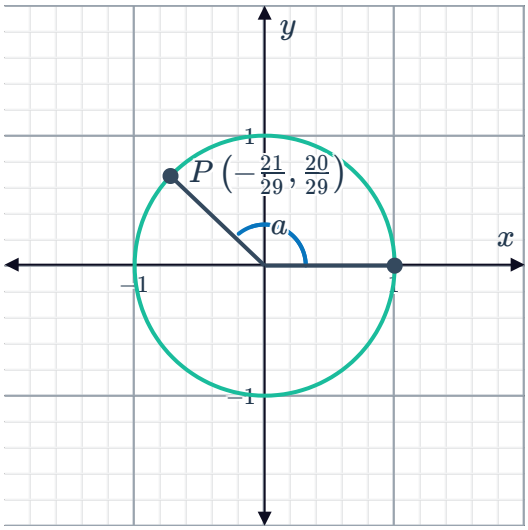
4 For each of the following graphs, find:

i $\sin a$ ii $\cos a$ iii $\tan a$

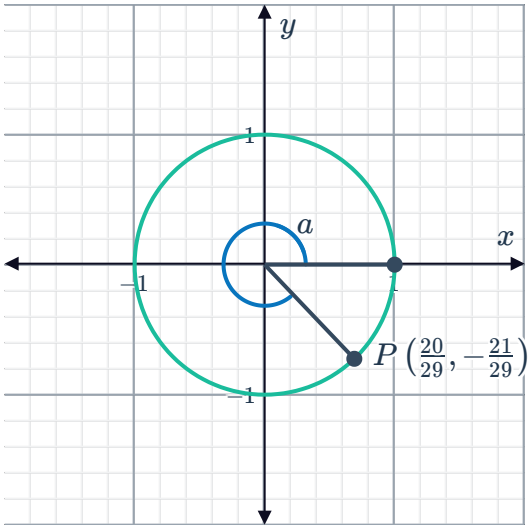
a



b



c



Identify quadrants

5 If $\cos\theta = \frac{-7}{8}$ and θ is in quadrant 3, find $\sin\theta$.

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6 State the quadrants in which θ could be located where θ is an angle of rotation corresponding to the point (x, y) on the unit circle such that:

a $\sin\theta > 0$ and $\cos\theta < 0$

b $\sin\theta < 0$ and $\cos\theta < 0$

c $\tan\theta < 0$ and $\cos\theta < 0$

d $\tan\theta > 0$ and $\sin\theta > 0$

e $\sin\theta < 0$ and $\csc\theta < 0$

f $\cos\theta > 0$ and $\sec\theta > 0$

g $\tan\theta > 0$ and $\cot\theta > 0$

h $\cos\theta < 0$ and $\sec\theta > 0$

i $\sec\theta > 0$ and $\csc\theta > 0$

j $\sec\theta > 0$ and $\csc\theta < 0$

k $\cot\theta > 0$ and $\csc\theta > 0$

l $\cot\theta > 0$ and $\sec\theta < 0$

7 State the quadrant where the point $P(x, y)$ is located if P is on the unit circle at a rotation of θ around the origin such that:

a $\sin\theta = \frac{5}{13}$ and $\cos\theta = \frac{12}{13}$

b $\sin\theta = -\frac{12}{37}$ and $\cos\theta = \frac{35}{37}$

c $\sin\theta = -\frac{20}{29}$ and $\cos\theta = -\frac{21}{29}$

d $\sin\theta = \frac{3}{5}$ and $\cos\theta = -\frac{4}{5}$

Equation of the terminal side of an angle

8 The equation of the terminal side of an angle θ in standard position is given by $x + 3y = 0$, for $x \geq 0$.

a Sketch the graph of the equation of the terminal side of θ.

b Using the point $(3, -1)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).

c Find the exact values of the following:

i $\sin\theta$ **ii** $\cos\theta$ **iii** $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

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9 The equation of the terminal side of an angle θ in standard position is given by $-3x + 2y = 0$, for $x \geq 0$.

a Sketch the graph of the equation of the terminal side of θ.

b Using the point $(2, 3)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).

c Find the exact values of the following:

i $\sin\theta$ **ii** $\cos\theta$ **iii** $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

10 The equation of the terminal side of an angle θ in standard position is given by $4x + y = 0$, for $x \leq 0$.

a Sketch the graph of the equation of the terminal side of θ.

b Using the point $(-1, 4)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).

c Find the exact values of the following:

i $\sin\theta$ **ii** $\cos\theta$ **iii** $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

11 The equation of the terminal side of an angle θ in standard position is given by $x - 4y = 0$, for $x \leq 0$.

a Sketch the graph of the equation of the terminal side of θ.

b Using the point $(-4, -1)$, find the exact side lengths of the right triangle formed by the terminal side of θ that was graphed in part (a).

c Find the exact values of the following:

i $\sin\theta$ **ii** $\cos\theta$ **iii** $\tan\theta$ **iv** $\csc\theta$

v $\sec\theta$ **vi** $\cot\theta$

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